General Physics – 1121TEIXB1S0290 2P

Introduction to Quantum Mechanics

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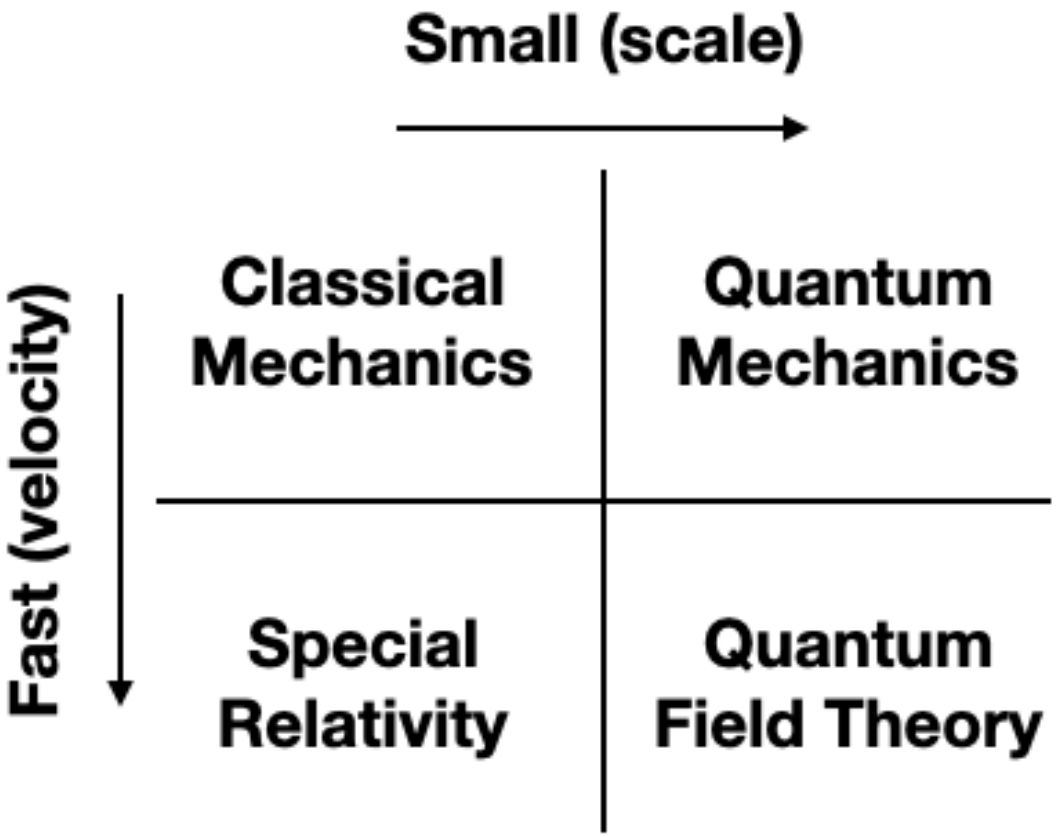
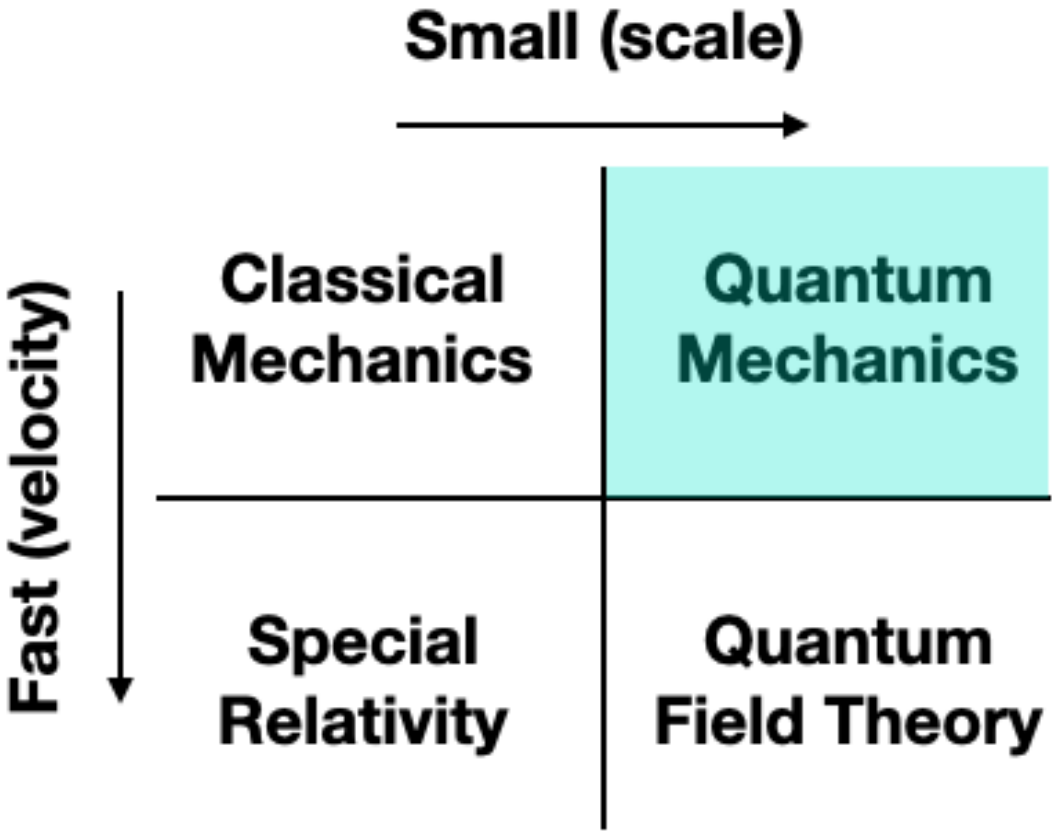
June 21, 2024@TKU

Physics with scale and velocity



Physics with scale and velocity

- Quantum physics:
  - A *non-relativistic* theory
  - It *usually* describes the small scale phenomenon.



Outline

1. Old Quantum Theory
2. Quantum Mechanics
3. Wave Function and Schrödinger Equation
4. Computational Basis States
5. Simple Qunantum Computation

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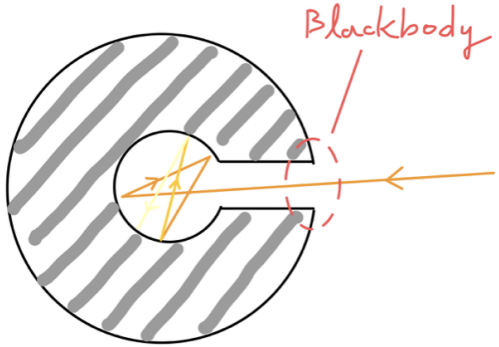
Quanta of Radiation

- Max Planck's calculational trick: Blackbody radiation as *quantum oscillators* $\varepsilon = hf$
  - The emitted/absorbed radiation energy

$$R(\lambda, T) = N(\lambda)\,\langle\varepsilon\rangle = \frac{2\pi hc^2}{\lambda^5(e^{hc/\lambda kT} - 1)} \qquad (1)$$

Boltzmann's constant $k = 1.38 \times 10^{-23}$ J/K
Planck's constant $h = 6.63 \times 10^{-34}$ J · s



Stefen-Bolztmann law
$P_{\text{blackbody}} = \sigma A T^4$

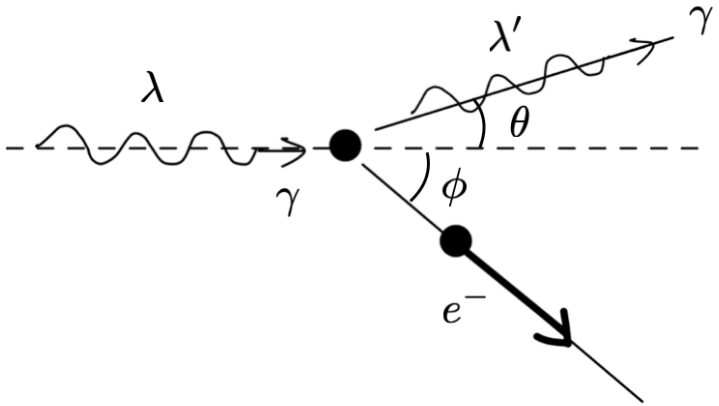
- Photoelectric effect: A. Einstein's *light-quantum* with $E_n = nhf$

$$\text{Single photon: } \underbrace{K_{\text{max}}}_{\text{Maximum kinetic energy of photoelectron}} = hf - \overbrace{\phi}^{\text{Work function: the threshold energy}} \implies \text{Threshold frequency } (K_{\text{max}} = 0) \text{ is } f_0 = \frac{\phi}{h}. \qquad (2)$$

- Compton scattering $\implies$ Light as particles.

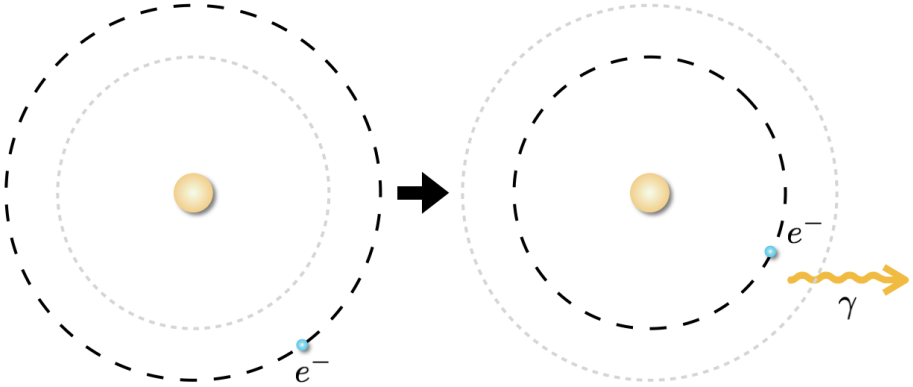
$$\Delta\lambda = \lambda' - \lambda = \frac{h}{m_e\, c}(1 - \cos\theta)$$

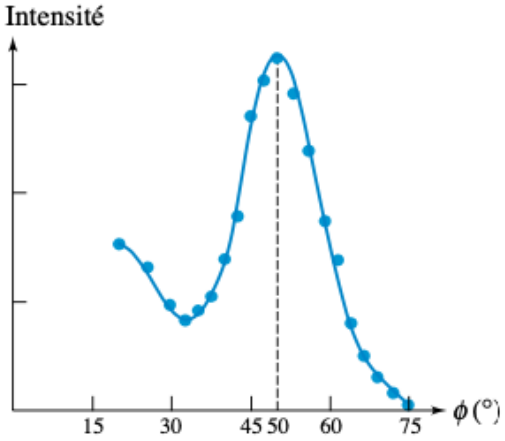
$$\approx 0.00243(1 - \cos\theta) \text{ nm.} \qquad (3)$$



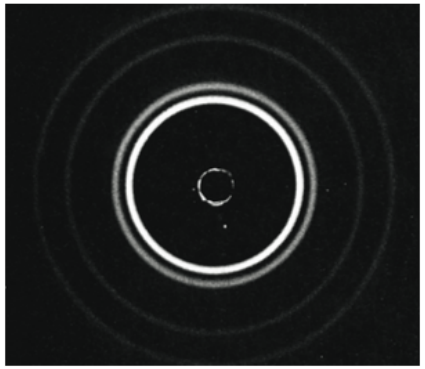
- J. J. Thomson's *plum pudding model*

Credit: The spruce eats – British Christmas Pudding (URL: `https://www.thespruceeats.com/traditional-christmas-pudding-recipe-435070`).





- Niels Bohr's atomic model
  - The electron only moves in certain circular orbit.
  - Transition of electron: Radiation with energy $hf = E_m - E_n$.
  - Quantization of angular mommentum $L = mvr = n\hbar$.
- The wavelength of de Broglie's *matter waves*: $\lambda = \frac{h}{p}$.
- *Electron-diffraction* experiments
  - Davisson-Germer experiment is similar to x-ray diffraction $\Longrightarrow$ Confirmation of the wave hypothesis.
  - G. P. Thomson's *diffraction ring* of electrons.



Credit: *Physique 3, 5th Edition*, Benson, etc., ERPI sciences, 2016.

Outline

1 Old Quantum Theory

2 Quantum Mechanics

3 Wave Function and Schrödinger Equation

4 Computational Basis States

5 Simple Qunantum Computation

Postulates of Quantum Mechanics

- Quantum equation of motion of *matter-waves*: *Schrödinger equation*

$$\boxed{i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t)} \text{ with the } \textit{reduced Planck's constant} \boxed{\hbar = \frac{h}{2\pi}.} \tag{4}$$

- *Superposition principle* of wave function

$$\boxed{\Psi(x,t) = \sum_i c_i\psi_i(x,t) = \sum_i c_i\mathcal{U}_i(t)\psi_i(x),} \quad \text{where} \quad \boxed{\mathcal{U}_i(t) = e^{-iE_it/\hbar}.} \tag{5}$$

- *Heisenberg's uncertainty principle*

$$\boxed{\Delta x\,\Delta p_x \geq \frac{\hbar}{2}} \quad \Longrightarrow \quad \Delta t\,\Delta E \geq \frac{\hbar}{2}. \tag{6}$$

- A physical observable $\mathcal{O}$ becomes a quantum *Hermitian* operator $\hat{\mathcal{O}} = \hat{\mathcal{O}}^\dagger = (\hat{\mathcal{O}}^*)^T$ which should be measured as a *expectation value* (the *statistical interpretation*):

$$\boxed{\mathcal{O} \longrightarrow \langle\hat{\mathcal{O}}\rangle = \frac{\langle\Psi|\hat{\mathcal{O}}|\Psi\rangle}{\langle\Psi|\Psi\rangle} = \frac{\int\Psi^\dagger\hat{\mathcal{O}}\Psi\,dx}{\int\Psi^\dagger\Psi\,dx},} \tag{7}$$

where $\langle\Psi_1|\Psi_2\rangle$ defines the *inner product* of two state vectors.

Note

A vector $\vec{A}$ can be expanded by

$$\vec{A} = A_x\,\vec{e}_x + A_y\,\vec{e}_y + A_z\,\vec{e}_z = \sum_i A_i\,\vec{e}_i \equiv \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = A_x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + A_y \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + A_z \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Quantum mechanically we have the identification of the *state vector* in the *Hilbert space* (*Dirac's notation*)

$$\vec{A} \longrightarrow \Psi \equiv |\Psi\rangle \quad \text{and} \quad \vec{e}_i \longrightarrow \psi_i \equiv \overbrace{|\psi_i\rangle}^{\text{ket vector}} \overset{\text{or}}{=} |i\rangle\,, \quad \underbrace{\psi_i^\dagger}_{\textit{hermitian conjugate} \text{ of } \psi_i} \equiv (|\psi_i\rangle)^\dagger = \overbrace{\langle\psi_i|}^{\text{bra vector}} \overset{\text{or}}{=} \langle i|\,. \tag{8}$$

So a state vector and its *hermitian conjugate* are

$$\Psi = \sum_i c_i \psi_i \equiv \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} \quad \text{and} \quad \Psi^\dagger = \sum_i \psi_i^\dagger\, c_i^* \equiv \begin{bmatrix} c_1^* & c_2^* & \cdots & c_n^* \end{bmatrix}. \tag{9}$$

Then the innert product of two basis vectors is

$$\vec{e}_i \cdot \vec{e}_j = \delta_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases} \longrightarrow \langle\psi_i|\psi_j\rangle = \int \psi_i^\dagger(x)\psi_j(x)\,dx = \delta_{ij}. \tag{10}$$

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Quantum Operator of Momentum and Energy

Given a *free* wave function $\Psi(x,t) = A e^{i(kx-\omega t)} \stackrel{\text{def}}{=} e^{-i\omega t}\Psi(x)$, then we have

$$\frac{\partial \Psi}{\partial x} = ik\, e^{i(kx-\omega t)} = ik\,\Psi, \tag{11a}$$

$$\frac{\partial \Psi}{\partial t} = -i\omega\, e^{i(kx-\omega t)} = -i\omega\,\Psi. \tag{11b}$$

Using the *de Broglie*'s matter-wave condition $\lambda = h/p$ and *Planck*'s quantum oscillator energy $E = hf$, we have new expression for momentum and energy

$$\left.\begin{aligned} p &= h/\lambda = (2\pi\hbar)/\lambda = \hbar k, \\ E &= hf = (2\pi\hbar)f = \hbar\omega. \end{aligned}\right\} \implies \Psi(x,t) = A e^{i(px-Et)/\hbar}. \tag{12a}$$

Quantum Operators

Multiply $\hbar$ to (11), the quantum mechanical momentum and energy operators, $\hat{p}$ and $\hat{H}$, are obtained

$$\hbar\frac{\partial \Psi}{\partial x} = ip\Psi \quad\implies\quad -i\hbar\frac{\partial}{\partial x}\Psi = p\Psi \quad\implies\quad \boxed{\hat{p} \stackrel{\text{def}}{=} -i\hbar\frac{\partial}{\partial x}.} \tag{13a}$$

$$\hbar\frac{\partial \Psi}{\partial t} = -iE\Psi \quad\implies\quad i\hbar\frac{\partial}{\partial t}\Psi = E\Psi \quad\implies\quad \boxed{\hat{H} \stackrel{\text{def}}{=} i\hbar\frac{\partial}{\partial t}.} \tag{13b}$$

$\implies$ The *quantization condition* of Heisenberg's *matrix mechanics*: $\boxed{\underbrace{[x, p_x]}_{xp_x - p_x x} = i\hbar.}$

Born's Statistical Interpretation

Born's *statistical interpretation* of quantum mechanics

- The wave function $\Psi(x,t)$ is called the *probability amplitude*.
- The *probability density function* is given by $P(x,t) = |\Psi(x,t)|^2 = \Psi^\dagger(x,t)\Psi(x,t)$.

$$\boxed{P(x,t)\,dx: \text{ the probability for finding the particle at position } x\text{, at time } t.} \tag{14}$$

- The conservation law of the probability (*continuity equation* of the probability)

$$\frac{\partial P(x,t)}{\partial t} + \vec{\nabla}\cdot\vec{J}(x,t) = 0 \text{ where } J = \frac{\hbar}{2m}\left[\psi^\dagger\frac{d\psi}{dx} - \psi\frac{d\psi^\dagger}{dx}\right] \text{ (}\textit{probability current}\text{).}$$

Note

A complex function $f(x)$ can be written two *real* functions $u(x)$ and $v(x)$

$$f(x) = u(x) + i\,v(x), \tag{15}$$

and its complex conjugate is

$$f^*(x) = \big(u(x) + i\,v(x)\big)^* = u^*(x) - i\,v^*(x) = u(x) - i\,v(x). \tag{16}$$

The square of its absolute value is defined by

$$|f(x)|^2 = f^*(x)f(x) = \big(u(x) - i\,v(x)\big)\big(u(x) + i\,v(x)\big) = u^2(x) + v^2(x). \tag{17}$$

- A *time-independent* (*stationay*) wave function can be written as

$$\Psi(x) \stackrel{\text{def}}{=} \Psi(x, t = 0) = \sum_i c_i \psi_i(x). \tag{18}$$

- The (continuous) *probability density function* is

$$P(x, t) = \Psi^\dagger(x, t)\Psi(x, t) = \Psi^\dagger(x) e^{+iEt/\hbar} e^{-iEt/\hbar} \Psi(x) = P(x). \tag{19}$$

- The (discrete) *probability mass function for each basis state* is

$$p_i = |c_i|^2 = c_i^* c_i, \tag{20}$$

  where c_i is called the *probability amplitude*.

- Given a *stationay* and *normalized* wave function $\Psi(x)$, the total probability should equal to one

$$\begin{aligned} 1 = \underbrace{\int P(x)\, dx}_{\text{continuous in } x} &= \int \left(\sum_i c_i \psi_i(x) \right)^\dagger \left(\sum_j c_j \psi_j(x) \right) dx \\ &= \sum_{i,j} c_i^* c_j \left(\underbrace{\int \psi_i^\dagger(x) \psi_j(x)\, dx}_{\delta_{ij}} \right) = \sum_j |c_j|^2 = \underbrace{\sum_j p_j}_{\text{discrete in } \psi_j}. \end{aligned} \tag{21}$$

Expectation Value

- The expectation value of the *continuous* variables x and x^2 are respectively

$$\langle x \rangle = \int \Psi^\dagger(x)\, x\, \Psi(x)\, dx = \int P(x) x\, dx, \tag{22a}$$

$$\langle x^2 \rangle = \int \Psi^\dagger(x)\, x^2\, \Psi(x)\, dx = \int P(x) x^2\, dx \tag{22b}$$

- The variation of x is Δx defined by its *standard deviation* σ_x:

$$\sigma_x^2 = \mathrm{Var}[x] = \langle x^2 \rangle - \langle x \rangle^2. \tag{23}$$

- The expectation value of p_x and p_x^2 are respectively

$$\langle p_x \rangle = \int \Psi^\dagger(x)\, \hat{p}_x\, \Psi(x)\, dx = \int \Psi^\dagger(x) \left(-i\hbar \frac{\partial}{\partial x} \Psi(x) \right) dx, \tag{24a}$$

$$\langle p_x^2 \rangle = \int \Psi^\dagger(x)\, \hat{p}_x^2\, \Psi(x)\, dx = \int \Psi^\dagger(x) \left(-\hbar^2 \frac{\partial^2}{\partial x^2} \Psi(x) \right) dx. \tag{24b}$$

- Similarly $\Delta p_x = \sigma_{p_x}$ is obtained by

$$\sigma_{p_x}^2 = \mathrm{Var}[p_x] = \langle p_x^2 \rangle - \langle p_x \rangle^2. \tag{25}$$

Then we can show Heisenberg's uncertainty principle

$$\Delta x\, \Delta p_x \geq \frac{\hbar}{2}. \tag{26}$$

Schrödinger Equation

- Classical mechanics: find the trajectory to satisfy the motion of

$$\text{Newton's equation} \quad \boxed{\vec{F} = \frac{d\vec{p}}{dt} = m\frac{d^2\vec{r}(t)}{dt^2} = -\vec{\nabla}U(x).} \tag{27}$$

- Quantum mechanics: find the wave function and *probability* to satisfy the motion of

$$\text{Schödinger equation} \quad \boxed{i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t),} \tag{28}$$

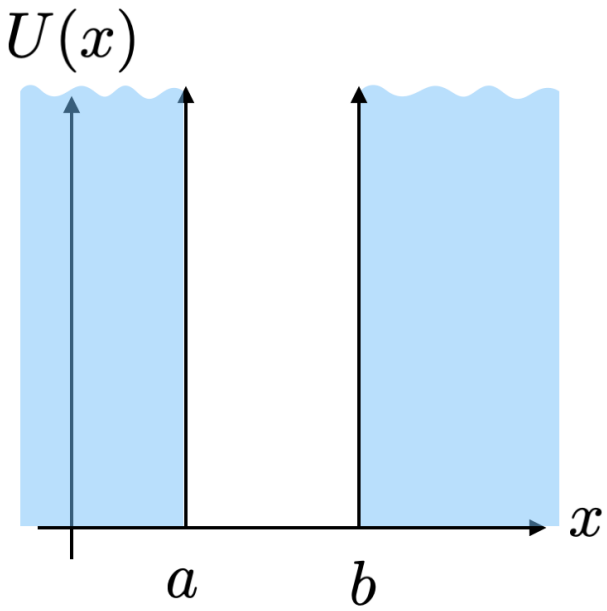
where $\hat{H} = \hat{K} + \hat{U}$ is the *hamiltonian* (the total energy) *operator* of the system.

A infinite potential well

$$\hat{K}(p_x) = \frac{\hat{p}_x^2}{2m} = \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} \tag{29}$$

and

$$U(x) = \begin{cases} \infty, & x < a, \\ 0, & a < x < b, \\ \infty, & x > b, \end{cases} . \tag{30}$$

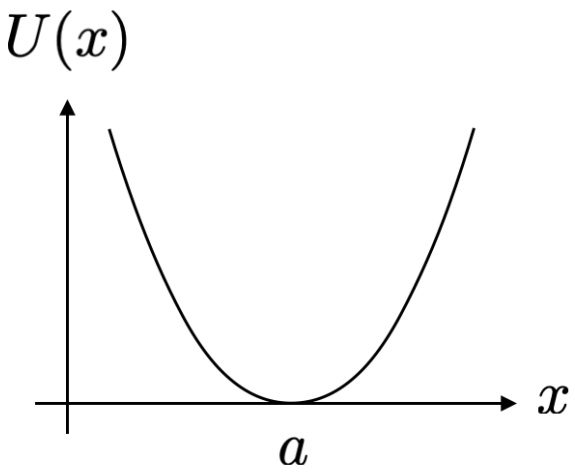


A simple harmonic oscillator

$$\hat{K}(p_x) = \frac{\hat{p}_x^2}{2m} = \frac{-\hbar^2}{2m}\frac{\partial^2}{\partial x^2} \tag{31}$$

and

$$U(x) = \frac{1}{2}k(x-a)^2. \tag{32}$$

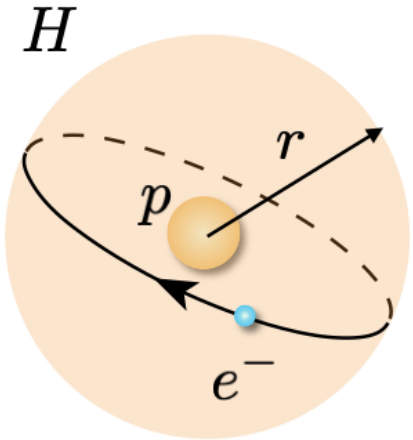


A hydrogen atom

$$\begin{aligned}\hat{K}(\vec{p}) &= \frac{\hat{p}^2}{2m} = \frac{-\hbar^2}{2m}\left(\hat{p}_r^2 + \hat{p}_\theta^2 + \hat{p}_\varphi^2\right) \\ &= \frac{-\hbar^2}{2m}\left\{\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}\right. \\ &\quad \left. + \frac{1}{r^2}\left(\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right)\right\}\end{aligned} \tag{33}$$

and

$$U(r) = -\frac{1}{4\pi\epsilon_0}\frac{kZe^2}{r^2} \quad (Z = 1 \text{ for hydrogen}). \tag{34}$$



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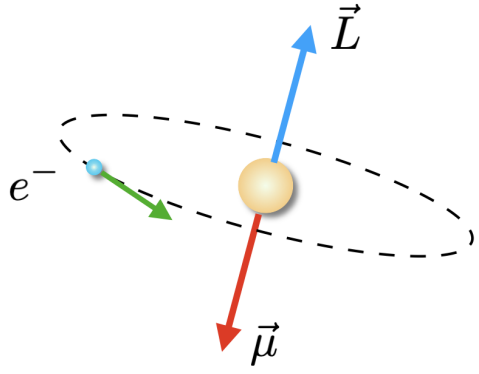
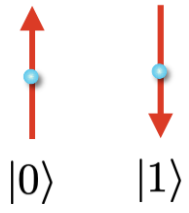
Quantum Bit

- For an electron, it has *spin* angular momentum $s = \frac{1}{2}$ (resp. *orbital* angular momentum $L = |\vec{r}||\vec{p}|\sin\theta$).
- Components: $m_s = \{-s, -s+1, \ldots, s-1, s\} = \pm\frac{1}{2}$ $\Longrightarrow$ A two-level system.
- For $m_s = +\frac{1}{2}$ and $m_s = -\frac{1}{2}$, the state functions are respectively written by

$$\chi_+ \longrightarrow |+\rangle \overset{\text{or}}{=} |\uparrow\rangle \overset{\text{or}}{=} |0\rangle \overset{\text{or}}{=} \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \tag{35a}$$

$$\chi_- \longrightarrow |-\rangle \overset{\text{or}}{=} |\downarrow\rangle \overset{\text{or}}{=} |1\rangle \overset{\text{or}}{=} \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{35b}$$

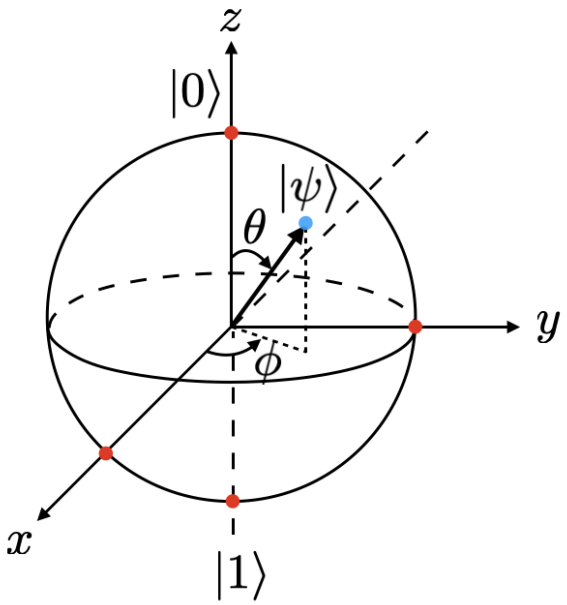




- A state is the superposition of these two *computational basis states*

$$\psi = \alpha\chi_+ + \beta\chi_- \longrightarrow |\psi\rangle = \alpha|0\rangle + \beta|1\rangle \overset{or}{=} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}. \tag{36}$$

- For a nomalized state, we have $|\alpha|^2 + |\beta|^2 = 1$.
- A Bloch sphere (an unit sphere)
  - A qubit state $|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$.
  - The linear combination of $|0\rangle$ and $|1\rangle$ states.



- Two *pure* states $\Longrightarrow$ A *two-particle* state $|\underbrace{\bullet}_{\text{1st state}}\ \underbrace{\bullet}_{\text{2nd state}}\rangle = \underbrace{|\bullet\rangle}_{\text{1st state}} \otimes \underbrace{|\bullet\rangle}_{\text{2nd state}}$:

$$\text{Four basis vectors:} \quad |00\rangle\,, \quad |01\rangle\,, \quad |10\rangle\,, \quad |11\rangle\,. \tag{37}$$

- Composite system with a composite states $\{|s, m_s\rangle\}$:

$$\left.\begin{array}{cc} |00\rangle & m_s = 1 \\ \frac{1}{\sqrt{2}}\Big(\,|01\rangle + |10\rangle\,\Big) & m_s = 0 \\ |11\rangle & m_s = -1 \end{array}\right\} \quad s = 1 \text{ (triplet)}, \tag{38a}$$

$$\frac{1}{\sqrt{2}}\Big(\,|01\rangle - |10\rangle\,\Big) \qquad m_s = 0 \qquad s = 0 \text{ (singlet)}. \tag{38b}$$

- A spin-*entangled* state (or Bell's state or EPR pairs): $|\psi\rangle \neq \underbrace{|\bullet\rangle}_{\text{1st state}} \otimes \underbrace{|\bullet\rangle}_{\text{2nd state}}$.

<u>*e.g.*</u>

$$|s = 1, m_s = 0\rangle = \frac{1}{\sqrt{2}}\Big(\,|01\rangle + |10\rangle\,\Big) \quad \text{and} \quad |s = 0, m_s = 0\rangle = \frac{1}{\sqrt{2}}\Big(\,|01\rangle - |10\rangle\,\Big). \tag{39}$$

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Quantum Logic Gate

X gate (quantum NOT gate):

- Classically $0 \longrightarrow 1$ and $1 \longrightarrow 0$.
- Quantum linear operation: $\alpha\,|0\rangle + \beta\,|1\rangle \longrightarrow \alpha\,|1\rangle + \beta\,|0\rangle$
  - The quantum NOT gate is definded by a 2×2 matrix

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{40}$$

  - If we write the stste as

$$|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \quad \Longrightarrow \quad |\psi'\rangle = X\,|\psi\rangle = X \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \beta \\ \alpha \end{bmatrix}. \tag{41}$$

Z gate:

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{42}$$

$Hadamard$ gate:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{43}$$

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Outline

6 Backup Slides

Relativistic Quantum Field Theory

Relativistic Theory

$$\text{QM} + \text{SR} + \underbrace{\text{Gauge Principles}}_{\text{Interaction \& conservation law}}.$$

- The relativistic equations of *Quantum Electrodynamics* (QED)
  $\Longrightarrow$ *Dirac equation* & *Maxwell equations*

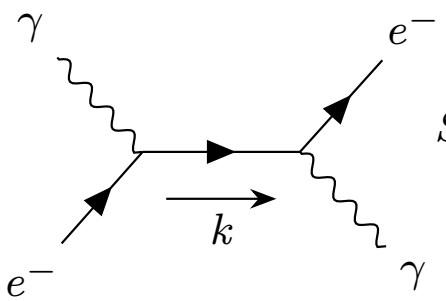
$$i\hbar c\gamma^\mu(\partial_\mu - ieA_\mu)\psi = mc^2\psi \longleftarrow \text{predict } \textit{antiparticles}, \tag{44a}$$

$$\partial_\mu F^{\mu\nu} = (4\pi/c^2)J^\nu, \tag{44b}$$

$$\text{where} \quad \begin{array}{c}\text{Electric current}\\ J^\mu = e\bar{\psi}\gamma^\mu\psi\end{array} \quad \text{and} \quad \begin{array}{c}\text{Electromagnetic fields } \vec{E} \text{ \& } \vec{B}\\ F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.\end{array} \tag{45}$$

with $A_\mu = (\phi, -\vec{A})$ the *electromagnetic potential*.

- The *leading order Feynman diagram* of the *compton scattering*



- The action of QED (in *nature unit* $\hbar = c = 1$)

$$S = \int d^4x \overbrace{\mathcal{L}}^{K-U} = \int d^4x \left\{\bar{\psi}i\gamma^\mu(\partial_\mu \overbrace{- ieA_\mu}^{\text{EM interaction}})\psi - m\bar{\psi}\psi - \frac{1}{4}F^{\mu\nu}F_{\mu\nu}\right\}.$$

- Quantization: $Z = \langle\text{vacuum}|\text{vacuum}\rangle = \int[d\psi][d\bar{\psi}][dA_\mu]\, e^{-iS}$.
- Vertex: $J^\mu A_\mu = (e\bar{\psi}\gamma^\mu\psi)A_\mu$ and $\partial_\mu J^\mu = 0$.